Quantizing the Closed String

Quantization

This process will be basically identical to what we did for the open string. However, this time I will be following Tong's notes so there will be more details and some notational differences.

Returning back to our original gauge, remember that the whole point of choosing such a gauge was to lock one world-sheet coordinate down. We can try to impose the original gauge conditions on our system but we run into an issue. We still have to deal with the periodicity introduced by the closed string. The first being time-dependent translations: $\sigma' = \sigma + s(\tau) \bmod l$

This can be locked into place with: $r_{\tau,\sigma}(\tau,0) = 0$

What does this mean? Well by imposing this extra gauge condition, we are basically setting the $\sigma = 0$ line orthogonal to τ.

Before this, your σ line could skew indefinitely due to periodicity. However, now the main $\sigma = 0$ line has been restricted not to skew, so it takes care of everything w.r.t the rest of the gauge conditions.

However, this does not take into account the purely spatial translations: $\sigma' = \sigma + s \bmod l$ but that's a DOF we keep for now.

With, this the procedure is almost exactly identical. The general solution will be given by:

$$x^i(\tau,\sigma) = x^i + \frac{p^i}{p^+}\tau + i\left(\frac{\alpha'}{2}\right)^{1/2} \sum_{n\neq 0}\left\{\frac{\alpha_n^i}{n}\exp\left[-\frac{2\pi i n(\sigma+c\tau)}{l}\right] + \frac{\tilde{\alpha}_n^i}{n}\exp\left[\frac{2\pi i n(\sigma-c\tau)}{l}\right]\right\}$$

We can see that the difference between the closed and open string is that the former permits left and right moving waves while the latter is constrained by BC. It also tells us that we can, in a closed string excite in a left or a right moving way, giving us more excitations.

This can be quantized now with:

$$[x^-, p^+] = i \; ; \; [x^i, p^j] = i\delta^{ij} \; ; \; [\alpha_m^i, \alpha_n^j] = m\delta^{ij}\delta_{m,-n} \; ; \; [\tilde{\alpha}_m^i, \tilde{\alpha}_n^j] = m\delta^{ij}\delta_{m,-n}$$

Instead of the ground state $|0,k\rangle$, we start with $|0,0;k\rangle$. We can create a general state then, by using:

$$|M,\tilde{M},k\rangle = \left[\prod_{i=2}^{D-1}\prod_{n=1}^{\infty}\frac{(\alpha^i_{-n})^{M_{in}}(\tilde{\alpha}^i_{-n})^{\tilde{M}_{in}}}{(n^{M_{in}}M_{in}!\,n^{\tilde{M}_{in}}\tilde{M}_{in}!)^{1/2}}\right]|0,0;k\rangle$$

The mass formula is then given by:

$$m^2 = 2p^+H - p^ip^i = \frac{2}{\alpha'}\left[\sum_{n=1}^{\infty}(\alpha^i_{-n}\alpha^i_n + \tilde{\alpha}^i_{-n}\tilde{\alpha}^i_n) + A + \tilde{A}\right] = \frac{2}{\alpha'}(M + \tilde{M} + A + \tilde{A})$$

We are now taking the zero-point energies of the right and left moving oscillators separately. This is honestly, pretty straightforward.

We can then sum all the zero-point energies to get: $A = \tilde{A} = \frac{2\cdot 0}{24}$

Now, what about that one gauge freedom we still have? Well this counts as a translation which is a symmetry associated with a Noether current (operator): $P = -\int_0^1 d\sigma\, \Pi^i \partial_\sigma X^i$

We can then plug in the mode expansion and do some fancy math (which I won't do) to get:

$$P = -\frac{2\pi}{l}\left[\sum_{n=1}^{\infty}(\alpha^i_{-n}\alpha^i_n - \tilde{\alpha}^i_{-n}\tilde{\alpha}^i_n) + A - \tilde{A}\right] = -\frac{2\pi}{l}(M - \tilde{M})$$

Basically, we can eliminate the degree of freedom by setting: $M - \tilde{M} = 0 \longrightarrow M = \tilde{M}$

There are always just as many right-moving waves as there are left-moving.